UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

PART A (10 MARKS)

Answer ALL questions.

1. How many times does a while loop execute when its condition is false initially?
   A. One
   B. Zero
   C. Until the user stops it
   D. Two

2. Which values can a bool variable hold?
   A. any decimal
   B. true or false
   C. any character
   D. any integer

3. Which tool represents an algorithm using standard graphical symbols?
   A. Debugger
   B. Compiler
   C. Flowchart
   D. Pseudocode

4. Which loop is commonly used to process every array element?
   A. for
   B. break
   C. if
   D. switch

5. Which term describes a finite sequence of precise steps that solves a problem?
   A. Compiler
   B. Variable
   C. Algorithm
   D. Array

6. Which keyword prevents a variable's value from being changed after initialisation?
   A. void
   B. break
   C. const
   D. static

7. Which statement immediately exits a loop?
   A. return 0
   B. continue
   C. break
   D. cin

8. What does the % operator return?
   A. The remainder of integer division
   B. The absolute value
   C. The square root
   D. The quotient

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9. Which input pattern stops when the user enters -1?
   A. A sentinel-controlled loop
   B. A function prototype
   C. A fixed for loop
   D. A switch statement

10. Which loop is guaranteed to execute its body at least once?
   A. do...while
   B. while
   C. if
   D. for

PART B (60 MARKS)

QUESTION 1

This question tests high-level languages and debugging.

a) Define syntax error and logic error.

(2 marks)

b) Explain TWO advantages of high-level languages over machine language.

(3 marks)

QUESTION 2

A program reads a positive integer N and calculates the sum and average of the numbers 1 to N.

a) Draw a flowchart for the sum and average calculation.

(8 marks)

b) Write pseudocode and a complete C++ program for the algorithm.

(5 marks)

QUESTION 3

This question tests operator precedence and assignment.

a) Evaluate the following expressions: (i) 7 + 3 * 2, (ii) (7 + 3) * 2, (iii) 17 % 5, (iv) 17 / 5, (v) 17.0 / 5.

(5 marks)

b) Write C++ statements to declare two integers a and b, read their values, and display their sum, difference, product and remainder.

(6 marks)

QUESTION 4

Study the following array program segment:

```
int a[5] = {2, 4, 6, 8, 10};
int s = 0;
for (int i = 0; i < 5; i++) {
   s += a[i];
}
cout << s;
```

a) Trace the program and state the output.

(6 marks)

b) Rewrite the segment using a do...while loop that sums only values greater than 5.

(9 marks)

QUESTION 5

A sports club stores sport names and medal counts in parallel arrays.

a) Declare two parallel arrays for five sports and initialise them with suitable values.

(8 marks)

b) Write a program segment to display each sport with its medals, then calculate the total and average medals.

(8 marks)

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PART C (30 MARKS)

QUESTION 1

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

QUESTION 2

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

END OF QUESTION PAPER